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Working with AI

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This session

- 5. Doing work with AI
- 6. Prompting tips
- 7. Permissions

5 · DOING WORK WITH AI

Where to work

Online chatbot

In a browser. Nothing to install.

Desktop app or CLI

Installed on your computer.
Works with files on your computer.

lab803.kerryback.com

A container on a server,
with the software already installed.

lab803

Sign in at lab803.kerryback.com with your NetID as the username and your Rice student ID (S followed by eight digits) as the password.

A container

Each of you has a container on a Linux server.

Claude Code and Python

Both are installed in each container.

Sonnet 5

Claude Code is connected to Sonnet 5.

Right panel

Claude Code. The prompt window is at the bottom.

Left panel

Files (the default),
Terminal, and View. Ignore
Terminal and View for now.

Three-dots menu in Files

Upload from your laptop
and download to your
laptop.

Workspace

Important

Refresh

Click Refresh, above the left panel, to see a new file in Files.

No Microsoft Office

Double-clicking a Word, PowerPoint, or Excel file will not open it. Office is not installed.

Download

Download Office files from the three-dots menu and open them on your laptop.

Ask

Ask a question, anything.
Type in the prompt window
and press Enter.

Upload

Upload a Word doc and ask
Claude to read it.

Download

Ask Claude to create a
Word doc, then download it
and read it.

Explore

Managing Claude

`Esc`

stop what it is doing, and keep the conversation

`Esc` twice

jump back to an earlier message and take a different path

`Ctrl-C` twice

kill Claude Code outright

`claude`

start it again, in a new conversation

Up arrow

your previous prompts

`Shift-Tab`

cycle the permission mode

`Esc` stops a wrong turn without throwing away the context you built. Two `Ctrl-C` s are for when the process itself is stuck. In lab803, press Enter after that to reconnect.

Resume and continue

Every conversation is written to disk in your container as you go.

WHAT YOU GET

`claude`

a new, empty conversation

`claude --continue`

the most recent one, where you left it

`claude --resume`

a list of your past conversations, to pick from

So the answer to a stuck terminal is always the same: two `Ctrl-C`s, then `claude --continue`.

Slash commands

Typed into Claude, not into the shell.

`/clear` wipe the conversation and start fresh without leaving

`/resume` switch to another conversation from inside this one

`/rename` name this conversation so you recognize it in that list

`/help` everything available

A conversation that has wandered sends all of that back to the model on every turn — you pay for it, and it crowds out the question you asked.

A few more worth knowing

`/context`

what is filling the context window right now

`/cost`

what this session has cost so far

`/model`

change models mid-conversation

`/permissions`

what is allowed without asking

`/init`

write a starting `CLAUDE.md` for this folder

`/mcp` · `/plugin`

connectors and plugins

Three prefixes work without a slash: `!` runs a shell command directly, `@` points at a file, and `#` adds a line to memory.

Start a new conversation for a new task

AI reads the entire conversation again each time you prompt. Irrelevant stuff consumes tokens and may be confusing.

When a conversation gets very long

Tell AI to write a handoff document so you can start a fresh conversation.

Managing conversations

Try it: build an Excel workbook

Tell Claude: Build an Excel workbook containing a mortgage amortization table. Include a chart of the amount going to interest and the amount going to principal each month.

Download and investigate

Download the file, open it on your own machine.

- 1 Click a payment cell. Formula, or a number?
- 2 Change the rate to 12%. Does the chart move?
- 3 Change the term to 15 years. Does the table shorten?

If any answer is no, say so in one sentence and ask Claude to fix that specifically.

- 1 Ask Claude to add a worksheet containing a plot of the monthly payment as a function of the interest rate.
- 2 Download and check.

Ask for something else

Breakout exercise

In lab803, the five Northwind spreadsheets are in `data/`. Elsewhere, download `northwind.zip`.

- 1 Ask: *Which countries generated the most revenue? Compute revenue as price times quantity.*
- 2 Ask: *Create a bar chart of revenue by country for the top five countries.*
- 3 Ask: *Create an Excel workbook containing the revenue table and the chart.*
- 4 Ask: *Create a Word document containing a one-page summary with the table and the chart.*
- 5 Ask: *Create a PowerPoint deck containing three slides: the question, the table, and the chart.*

At lab803, use Refresh and then download files to view them.

Claude and ChatGPT in Excel

CLAUDE FOR EXCEL

From Anthropic. Paid Claude plans: Pro, Max, Team, Enterprise.

Cites the cells behind its answers. Edits cells without breaking formulas. Builds pivot tables, sorts, and filters.

Also available for PowerPoint, Word, and Outlook.

CHATGPT FOR EXCEL

From OpenAI. All ChatGPT plans. Also works in Google Sheets.

Builds sheets from a description, writes formulas, and sorts and updates tables.

Asks before it edits the workbook.

6 · PROMPTING TIPS

Plan

Prompt

Think about the best way to do this, formulate a plan, and tell me what your plan is.

Afterwards

Prompt

Explain what you did step by step. Did you make any decisions that we have not discussed?
Write this as an HTML doc.

HTML is easy for AI to write and easy to read. Double-click an HTML file in File Explorer (Finder on Mac) and it will open in your browser.

Numbers need code

If a number appears in prose and no code produced it, it was predicted.

Prompt

```
Compute this in Python and show me the code and the output. Do not report a number you did not compute.
```

The same rule applies to counts, percentages, dates, and anything summed across rows.

Verify claims

Ask what would falsify it

“What would have to be true for this to be wrong, and how would I check?”

Make it show its source

The file, the row count, the line of code.

Generator-critic

- 1 Get the first answer.
- 2 Start a fresh conversation and give it the answer without the reasoning that produced it.
- 3 Ask it to find what is wrong with it.
- 4 Take the objections back to the first conversation.

The second conversation has no stake in the first answer.

Most ‘prompt engineering’ techniques from a few years ago are now either useless or actually counterproductive. Don’t tell it “you are an expert data scientist” or “think step by step.”

Tell it what you want, not what to do

I want ... How can we do it?

If something doesn’t work

Try again.

Treat AI as an infinitely patient colleague with a poor memory.

Good enough prompting

AI is a prediction machine, not a rational person

Suppose an AI agent has told you in the path that Method A beats Method B for some task. You ask it to do the task.

Do not assume it will use what it knows. If “A beats B” is not in the context window, and B appeared more often in its training data, it will probably use B.

This is the value of “plan before you act.” Planning surfaces “A beats B,” leading Claude to select A for the task.

Try it: planning

Three rigs, twenty wells, minimize lost production: [workover.md](#). At lab803 it is in `data/`.

- 1 Upload `workover.md` and ask: *Find the best routes for the rigs.*
- 2 Start a new conversation (`/clear` in the Claude tab). Upload the file again and ask: *Plan how to do this and share your plan.* Read the plan, then ask it to proceed.

Did it use the same method both times?

Try it: explain and critique

- 1 Ask: *Write an HTML doc explaining what you did, step by step.*
- 2 Start a new conversation. Tell it to read the HTML doc and ask: *Read this and critique the method.*

7 · PERMISSIONS

What Claude asks before it acts

MODE	WHAT IT DOES
Manual	asks before the first use of each tool
Plan	reads and explores, edits nothing
Accept edits	file edits go through without asking
Auto	approves on its own, with a check on each action
Don't ask	refuses anything it would otherwise ask about
Bypass	no prompts at all

Enforced by the program around the model, not by the model. A line in `CLAUDE.md` shapes what Claude tries to do; it does not change what is allowed.

Permissions in Claude Desktop

The desktop app runs Claude Code, so the modes are the same and carry the same names.

What it can reach

the folders you connect to the conversation, and nothing else

What it asks about

the first use of each tool, in Manual mode

What “don’t ask again” does

writes a rule that outlives the conversation

Granting a folder is the decision that matters. The mode decides how often you are asked; the folder decides what can be reached at all.

Everything is allowed already, in `~/.claude/settings.json`:

```
1 "permissions": {  
2   "allow": ["Bash(*)", "Read(*)", "Write(*)", "Edit(*)",  
3           "Glob(*)", "Grep(*)", "WebFetch(*)", "WebSearch(*)"],  
4   "deny": []  
5 }
```

So nothing prompts you all week. That is a decision about this container — it holds your course files and is reachable only by you — not about how to run an agent on your own machine.

Permissions in your container